

All practice questions and explanations for this topic taken from:

LIPPINCOTT'S ILLUSTRATED Q&A REVIEW OF HISTOLOGY

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Edition:

1ST ED.

1. A 6-year-old boy who does not respond when his mother calls him is brought in for evaluation. Large earwax masses are found to be plugging both of his ears. The major component of earwax, cerumen, is produced by which of the following structures in the external acoustic meatus?
 - a. Cartilage tissue
 - b. Eccrine sweat gland
 - c. Modified apocrine sweat gland
 - d. Hair follicle
 - e. Tympanic membrane

2. The mother of an 18-month-old girl complains that her daughter has a fever, cough, and nasal discharge. Physical examination reveals an upper respiratory tract infection and a bulging tympanic membrane. The patient is subsequently diagnosed with otitis media secondary to an upper respiratory tract infection. The infection and inflammation in this patient affect which of the following areas of the ear?
 - A. Cochlear labyrinth
 - B. External acoustic meatus
 - C. Semicircular canal
 - D. Tympanic cavity
 - E. Vestibular labyrinth

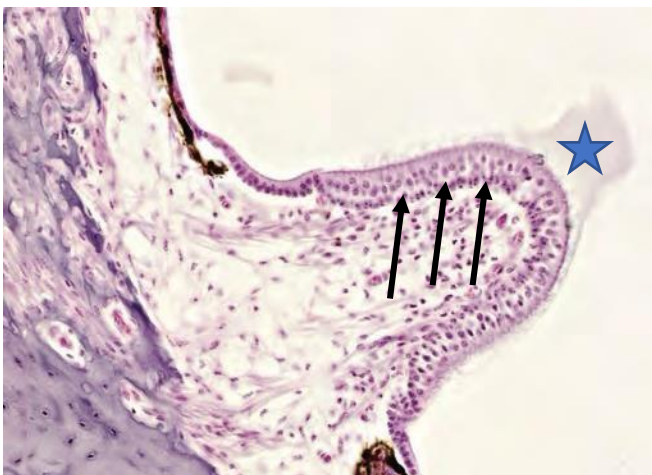
3. A 36-year-old woman complains about progressive hearing loss in the last year and is diagnosed with otosclerosis. Otosclerosis primarily affects which of the following structures of the ear, leading to impaired hearing?
 - A. Auditory ossicles
 - B. Auditory tube
 - C. Cochlear labyrinth
 - D. Tympanic membrane
 - E. Vestibular labyrinth

4. A section of the internal ear obtained from a guinea pig is examined in the histology laboratory (shown in the image). The space indicated by the asterisk represents which of the following structures?



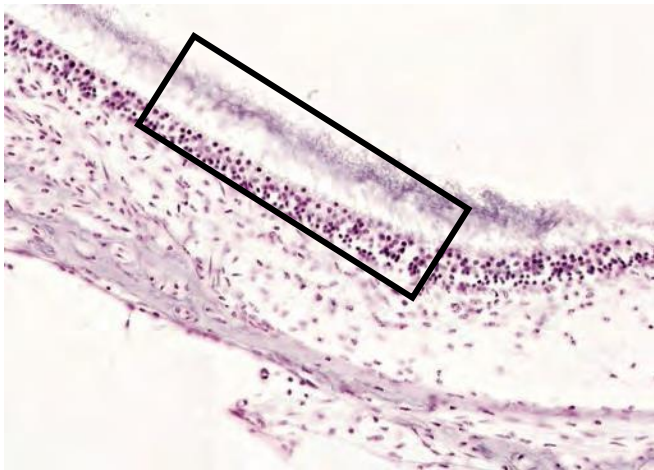
- (A) Ampulla of a bony semicircular canal
- (B) Cochlea
- (C) Internal acoustic meatus
- (D) Membranous ampulla of a semicircular duct
- (E) Tympanic cavity

5. The ampulla of the semicircular duct described in Question 4 is examined at high magnification (shown in the image). Identify the cells indicated by the arrows.



- (A) Afferent neurons
- (B) Schwann cells
- (C) Supporting cells
- (D) Type I hair cells
- (E) Type II hair cells

6. An epithelial thickening is found in another region of the membranous labyrinth (indicated in the rectangular box, shown in the image). Which of the following terms describes histological features of this epithelial receptor?



- (A) Crista ampullaris
- (B) Macula
- (C) Organ of Corti
- (D) Saccule
- (E) Utricle

7. An 18-year-old man experiences seasickness while on a deep-sea fishing excursion. Overstimulation of which of the following structures causes vertigo in this patient?

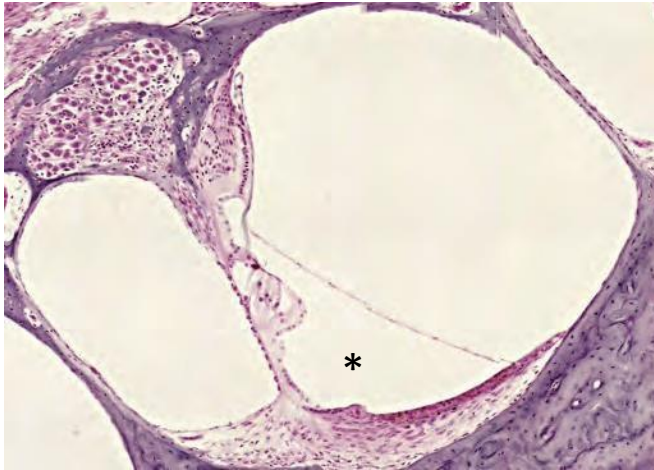
- (A) Cochlear labyrinth
- (B) Crista ampullaris
- (C) Macula
- (D) Organ of Corti
- (E) Semicircular ducts

8. Cells in which of the following structures produce endolymph in the scala media?

- (A) Basilar membrane
- (B) Organ of Corti
- (C) Periosteum

- (D) Stria vascularis
- (E) Vestibular membrane

9. A section of the cochlear canal is examined at higher magnification (shown in the image). Identify the space indicated by the asterisk.



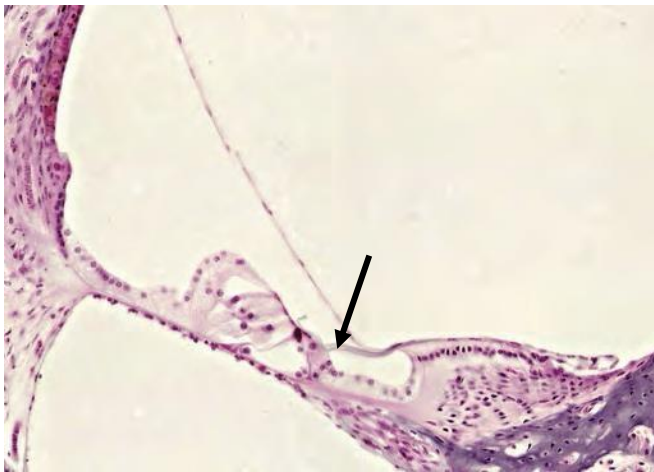
- (A) Cochlear canal
- (B) Internal spiral tunnel
- (C) Scala media
- (D) Scala tympani
- (E) Scala vestibule

10. The scala media at the base of the cochlea specimen described in Question 9 is examined at higher magnification (shown in the image). Identify the structure indicated by the arrow.



- (A) Basilar membrane
- (B) Osseous spiral lamina
- (C) Spiral ligament
- (D) Tectorial membrane
- (E) Vestibular membrane

11. A section through the cochlea near the apex is examined in the histology laboratory (shown in the image). Identify the structure indicated by the arrow.



- (A) Basilar membrane
- (B) Osseous spiral lamina
- (C) Spiral limbus
- (D) Tectorial membrane
- (E) Vestibular membrane

12. Which of the following structures transforms vibrations of the ossicles into waves of fluid in the cochlea?

- (a) Ampullae of semicircular ducts
- (b) Oval window
- (c) Round window
- (d) Saccule
- (e) Utricle

13. You are invited to attend an outdoor concert in the park during the summer. Your ear detects delightful changes in the frequency of sound vibrations (notes with different

pitch). Which of the following structures in the internal ear encodes this frequency-based acoustic information and converts it into nerve impulses that are conveyed to the brain?

- (A.) Basilar membrane
- (B.) Crista ampullaris
- (C.) Macula of saccule
- (D.) Macula of utricle
- (E.) Vestibular membrane

14. Which of the following types of cells initiates neuronal transduction during sound perception?

- (A.) Hair cells of the crista ampullaris
- (B.) Hair cells of the macula of utricle
- (C.) Hair cells of the spiral organ of Corti
- (D.) Phalangeal cells
- (E.) Pillar cells

Answers

1. **The answer is C: Modified apocrine sweat gland.** Modified apocrine glands are found in the skin lining the external acoustic meatus of the external ear. They are also referred to as ceruminous glands, because the cells secrete cerumen—a waxy, yellowish material. The ear consists of three chambers, from lateral to medial: external ear, middle ear, and internal ear. The external acoustic meatus is a skin-lined canal through the temporal bone extending from the auricle to the tympanic membrane. The skin here contains sebaceous glands and hair follicles, but lacks eccrine sweat glands (choice B). Earwax is the mixture of secretions of the ceruminous glands and sebaceous glands. It has important protective and antimicrobial functions. Excessive accumulation of cerumen can block the meatus and result in conducting hearing loss. Other listed structures do not produce cerumen. **Keywords:** Ears, external acoustic meatus, ceruminous glands
2. **The answer is D: Tympanic cavity.** Tympanic cavity is an air-filled space within the temporal bone that is separated from the external acoustic meatus laterally by the tympanic membrane and from the internal ear medially by the bony wall. The tympanic cavity, along with three small auditory ossicles, is referred as the middle ear. The auditory tube (eustachian tube or pharyngotympanic tube) connects the middle ear to the nasopharynx. Otitis media is very common in childhood. Upper respiratory tract infections can cause swelling and obstruction of the auditory tube. Air in the middle ear cannot enter the tube and is absorbed through the mucosa, leading to negative pressure in the middle ear cavity. Accumulation of extravascular fluid results in a middle ear effusion. Semicircular canals, cochlear

labyrinth, and vestibular labyrinth (choices C, A, and E) are structures located in the internal ear and are not involved in otitis media. **Keywords:** Otitis media, tympanic cavity

3. **The answer is A: Auditory ossicles.** The auditory ossicles are three small bones that are articulated with one another in the tympanic cavity. They include, from lateral to medial, the malleus, incus, and stapes. The malleus attaches to the tympanic membrane and the stapes to the membrane of the oval window (one of two membrane-covered openings of the medial bony wall of the middle ear). The incus links the malleus and the stapes. Sound waves reach the tympanic membrane through the external acoustic meatus and cause vibration of the latter. The three ossicles convert air vibration into mechanical vibration in the fluid-filled chambers of the internal ear through the oval window. In the internal ear, sensory receptors convert mechanical forces into electrical impulses. Otosclerosis is an autosomal dominant hereditary defect uniquely affecting the temporal bone and the auditory ossicles. It is the most common cause of acquired conductive hearing loss in young and middle-aged adults in the United States. The most frequently affected site is the stapes, immediately anterior to the oval window. With progressive immobilization of the footplate, the stapes can no longer vibrate and transmit sound waves into the internal ear. **Keywords:** Otosclerosis, auditory ossicles
4. **The answer is D: Membranous ampulla of a semicircular duct.** The internal ear, located within the temporal bone, consists of the bony labyrinth and the membranous labyrinth. The membranous labyrinth represents a set of continuous epithelium-lined tubes and chambers filled with endolymph. Membranous labyrinth is suspended in the perilymph of the larger bony labyrinth—a set of interconnected canals and spaces. Three semicircular ducts of the membranous labyrinth, situated, respectively, within the three semicircular canals, are oriented at approximately right angles to each other in sagittal, frontal, and horizontal planes. They open into a central oval space called the vestibule. The end of each semicircular canal and duct is dilated to form the ampulla. As seen in the image, the crista ampullaris (indicated by the asterix) is located in the membranous ampullae of the semicircular ducts; this structure provides a sensory receptor for angular movements of the head. Ampulla of a bony semicircular canal (choice A) is the space surrounding the membranous ampulla. Cochlea (choice B) refers to the cone-shaped helical space connected with and anterior to the vestibule. Internal acoustic meatus (choice C) is a channel in the temporal bone for the vestibulocochlear nerve to pass. Tympanic cavity (choice E) is not seen in this image. **Keywords:** Ears, semicircular duct, bony labyrinth, membranous labyrinth
5. **The answer is C: Supporting cells.** The enlarged ampulla end of each semicircular duct contains a crista ampullaris, a mechanoreceptor formed by an

elongated ridge-like thickening of the epithelium. The ridge is perpendicular to the long axis of the semicircular duct. Crista ampullaris consists of hair cells, supporting cells, and nerve endings associated with hair cells. Cell types can be distinguished based on the location of the cells. The supporting cells are basal, while the hair cells are more superficial. There are two types of hair cells (choices D and E) in the epithelium, but they cannot be distinguished by light microscopy. Electric impulses generated by hair cells are transmitted by afferent nerve fibers. Afferent neuron cell bodies (choice A) and Schwann cells (choice B) are not located in the crista ampulla. A gelatinous proteoglycan layer known as the cupula (indicated by the star) is seen covering the crista ampullaris.

Keywords: Ears, crista ampullaris

6. **The answer is B: Macula.** Maculae represent small areas of thickened neuroepithelial cells in the saccule and the utricle that sense the position of the head and its linear movement. Three semicircular ducts and two connected sacs (the saccule and utricle) comprise the vestibular labyrinth. The saccule and utricle (choices D and E) are located within the central vestibule and are lined by a simple squamous epithelium. Two maculae are found in the walls of the saccule and utricle, and they are oriented perpendicular to each other. Histologically, they are composed of mechanosensitive hair cells, supporting cells, and associated nerve endings. A layer of gelatinous polysaccharide material (otolithic membrane) covers the surface of macula that faces the endolymph. The outer surface of the otolithic membrane contains otoliths, small crystalline bodies of calcium carbonate. Maculae respond to linear acceleration, gravity, and tilting of the head. Crista ampullaris (choice A) is located in the ampulla of the semicircular ducts. Organ of Corti (choice C) is found in the cochlea.

Keywords: Ears, macula, saccule, utricle

7. **The answer is C: Macula.** The stereocilia of the hair cells are embedded in the gelatinous otolithic membrane covering the maculae are sensitive to gravity and linear acceleration. Otoliths lie on the outer surface of the otolithic membrane and are heavier than the endolymph. When the head is not moving, the stereocilia bundles are deflected by gravity. When the individual is moving in a straight line, the otolithic membrane with the stereocilia of the hair cells is dragged by inertia, and this movement activates and depolarizes the hair cells. When one is moving by flight, train, or ship, repetitive changes in linear acceleration and direction can overstimulate the maculae in the utricles and cause motion sickness. Spinning the body and rotational movements create excessive stimulation of the crista ampullaris and semicircular ducts (choices B and E), which also cause vertigo. Organ of Corti (choice D) is situated in the cochlear labyrinth (choice A) and is responsible for hearing perception. **Keywords:** Motion sickness

8. **The answer is D: Stria vascularis.** Bordering the lateral aspect of the scala media the periosteal tissue adhering to the inner surface of the bony cochlea is thickened. This thickened periosteal connective tissue is referred as the spiral

ligament. Its endolymph facing surface is lined by a specialized secretory epithelium (stria vascularis) that contains a complex capillary network and produces most of the endolymph in the membranous labyrinth. There are three cell types in the stria vascularis, namely (1) marginal cells lining the endolymphatic space of the scala media, (2) pigment containing cells scattered among the capillaries, and (3) basal cells separating the stria vascularis from the underlying connective tissue of the spiral ligament. Cells of periosteum (choice C) and cells lining the perilymph facing surface of the membranous labyrinth secrete the perilymph that fills the scala vestibuli and scala tympani. Vestibular membrane (choice E), also termed the Reissner membrane, is a thin membrane composed of two layers of simple squamous epithelium separated by a basal lamina. It serves mainly as a diffusion barrier between endolymph in the scala media and perilymph in the scala vestibule. None of the other structures produce endolymph.

Keywords: Ears, endolymph, stria vascularis

9. **The answer is C: Scala media.** The cochlear duct, the membranous cochlear labyrinth, is suspended in the cochlear canal (bony cochlea, choice A). The cochlear canal is divided into three parallel longitudinal channels (compartments) running from the base to the apex of the cochlea. The cochlear duct itself is the triangular middle compartment (scala media) that contains endolymph. It is bounded superiorly by the vestibular membrane and inferiorly by the basilar membrane with the organ of Corti (indicated by the arrow) resting upon it. Laterally, the scala media is bordered by a special epithelium termed stria vascularis (indicated by the arrowhead). Scala vestibule and scala tympani (choices D and E) are perilymph containing channels flanking the scala media. The scala vestibule begins at the oval window, and the scala tympani ends at the round window membrane (second tympanic membrane). They communicate with each other at the apex of the cochlea, through a narrow slit (helicotrema). Internal spiral tunnel (choice B) is a space within the organ of Corti. **Keywords:** Ears, cochlea, scala media

10. **The answer is A: Basilar membrane.** The basilar membrane is a thick basement membrane stretching from the spiral ligament (choice C) laterally to the osseous spiral lamina (choice B) medially. It forms the floor of the scala media and separates it from the scala tympani. The highly specialized epithelium, the spiral organ of Corti, superimposes on the basilar membrane. A layer of cuboidal cells secreting perilymph lines its surface facing the scala tympani. The width and stiffness of the basilar membrane vary as it coils from base to apex of the cochlea. It is widest and least stiff at the apex of the cochlea and narrowest and most stiff at the base. Sound waves transmitted into the inner ear induce movement of fluid in the cochlea that causes displacement of the basilar membrane. Hair cells of the organ of Corti are subsequently stimulated and activated to convert these mechanical signals into electric nerve impulses.

Keywords: Ears, basilar membrane

11. **The answer is D: Tectorial membrane.** The tectorial membrane is a stiff, gelatinous acellular plate that extends from the spiral limbus. The spiral limbus represents a thickened periosteum of the osseous spiral lamina on the medial aspect of the scala media. The lateral margin of the tectorial membrane overlies the organ of Corti. The stereocilia of the outer hair cells in the organ of Corti are embedded in the lower surface of the tectorial membrane. The tectorial membrane is composed of collagen types II, V, and IX in an amorphous ground substance containing tectorin and otogelin; these glycoproteins are unique to the inner ear and are secreted by cells in the spiral limbus. None of the other structures exhibit characteristic features of the tectorial membrane.

Keywords: Ears, tectorial membrane

12. **The answer is B: Oval window.** The oval window and round window are two openings of the bony labyrinths within the temporal bone. The oval window is situated on the lateral wall of the vestibule of the bony labyrinth. The footplate of the stapes fits into the oval window. On the inner side, the oval window is the beginning of the scala vestibuli. The three-ossicle chain connects the tympanic membrane to the oval window. Movement of the stapes induced by the vibration of the tympanic membrane stirs up the mechanical vibration of the perilymph contained in the scala vestibuli, which in turn causes vibration of the endolymph in the scala media and, subsequently, the perilymph in the scala tympani. The round window (choice C) is located at the inferior aspect of the base of the cochlea and is covered by an elastic membrane termed secondary tympanic membrane. Pressure changes of fluid in the cochlea cause movement (bulging out or in) of this membrane. None of the other structures mediate sound wave conduction from the middle ear to the internal ear.

Keywords: Sound conduction, ears, oval window

13. **The answer is A: Basilar membrane.** As sound vibrations are transferred to the internal ear, a pressure pulse of the perilymph of the scala vestibule causes a traveling wave of deformation along the basilar membrane. The traveling wave of sound of a specific frequency reaches its peak amplitude at a particular location along the basilar membrane. As discussed earlier, the basilar membrane is narrow and relatively stiff at the base of the cochlea but increases in width and decreases in stiffness as it coils toward the apex of the cochlea. High frequency sounds cause maximal amplitude of the basilar membrane near the base of the cochlea. By contrast, the basilar membrane near the apex of the cochlea undergoes maximal displacement in response to low frequency sounds. Thus, different sites along the basilar membrane are specific for sounds with particular frequencies (pitch) and provide a structural basis for frequency discrimination. The receptor cells of the organ of Corti resting on a particular site of the basilar membrane respond best to sounds at particular frequency and convert the mechanical tuning of the basilar membrane into nerve pulses. The degree of displacement of the

basilar membrane, in other words, the amplitude at any particular frequency, reflects the intensity or loudness of sound. None of the other structures encode acoustic information based on sound frequency or amplitude. Keywords: Ears, basilar membrane

14. **The answer is C: Hair cells of the spiral organ of Corti.** The receptor hair cells of the organ of Corti are supported and surrounded by phalangeal cells. At their apical surface, stereocilia of the hair cells attach to the tectorial membrane. Thus, stereocilia connect the basilar membrane to the tectorial membrane. The basilar membrane stretches from the osseous spiral lamina medially to the lateral spiral ligament, whereas the tectorial membrane hinges from the spiral limbus. Vibrations of the basilar membrane and tectorial membrane create a shearing effect that deflects and activates stereocilia of the hair cells. The activated hair cells generate action potentials that are conveyed by the cochlear nerve to the central nervous system. Hair cells of the crista ampullaris and macula (choices A and B) are receptor cells responsible for balance and equilibrium. Phalangeal and pillar cells (choices D and E) are support cells.